Microsoft Excel for
Data & Business Analysis

Course Plan

Pre-Class & Post-Class Activities | Overview of In-class topics

DAY 1 — Introduction to Excel & Spreadsheet Fundamentals

Objective: *Understand what Excel is, navigate the interface confidently, and perform basic data entry, formatting, and simple formulas.*

PRE-CLASS (Self Study)	IN-CLASS (Topics We Will Cover)	POST-CLASS (Assignment)
• What is Microsoft Excel? (read a beginner article) • Understand: Rows, Columns, Cells & Worksheets concept • Install Microsoft Excel or access Excel Online (free via Microsoft 365 account) • Create a free Microsoft account for Excel Online	• Introduction to Excel – What & Why • Excel Interface Tour – Ribbon, Quick Access Toolbar, Formula Bar • Workbooks vs Worksheets vs Cells • Cell References – A1 Notation • Navigating Excel – Keyboard Shortcuts (Ctrl+Arrow, Ctrl+Home/End) • Data Entry – Text, Numbers, Dates, Currency • AutoFill & Flash Fill • Basic Formatting – Bold, Italic, Underline, Font Size & Color • Cell Alignment – Left, Center, Right, Merge & Center • Number Formatting – General, Number, Currency, Percentage, Date • Column Width & Row Height Adjustments • Inserting & Deleting Rows/Columns • Freeze Panes & Split View • Basic Formulas – =SUM(), =AVERAGE(), =MIN(), =MAX(), =COUNT() • Introduction to AutoSum • Saving Files – .xlsx, .csv formats • Printing Basics – Print Preview, Page Layout	• Create a Personal Monthly Budget spreadsheet with income and expenses • Format the budget with colors, borders, and currency formatting • Use SUM and AVERAGE formulas to calculate totals and averages • Practice: Enter a dataset of 20 students with Name, Age, Marks columns • AutoFill exercise: Generate numbers 1–50 using AutoFill • Flash Fill exercise: Split full names into First and Last name columns • Mini Project: Create a Class Attendance Tracker – mark Present/Absent for 10 students over 5 days

DAY 2 — Formulas, Functions & Data Management

Objective: *Master essential Excel functions for calculations, text manipulation, and logical decisions, and manage data effectively with tables and sorting.*

PRE-CLASS (Self Study)

- Revise: Basic formulas from Day 1 (SUM, AVERAGE, MIN, MAX)
- Read: Difference between Absolute and Relative Cell References
- Get an overview of Excel's IF function
- Read: What is an Excel Table (ListObject)?

IN-CLASS (Topics We Will Cover)

- Cell References Revisited – Relative (A1), Absolute (A1), Mixed ($A1 / A$1)
- Referencing Across Sheets – Sheet2!A1
- Logical Functions – IF(), AND(), OR(), NOT(), IFS()
- Nested IF Statements
- Text Functions – CONCATENATE(), CONCAT(), TEXTJOIN(), LEN(), LEFT(), RIGHT(), MID()
- Text Functions – UPPER(), LOWER(), PROPER(), TRIM(), SUBSTITUTE(), FIND()
- Date & Time Functions – TODAY(), NOW(), DATE(), YEAR(), MONTH(), DAY()
- Date Functions – DATEDIF(), EDATE(), EOMONTH(), NETWORKDAYS()
- Lookup Functions – VLOOKUP(), HLOOKUP()
- Modern Lookups – XLOOKUP(), INDEX(), MATCH()
- Statistical Functions – COUNTIF(), COUNTIFS(), SUMIF(), SUMIFS(), AVERAGEIF()
- Rounding – ROUND(), ROUNDUP(), ROUNDDOWN(), CEILING(), FLOOR()
- Math Functions – MOD(), INT(), ABS(), POWER(), SQRT()
- Excel Tables – Creating, Naming, Structured References
- Sorting Data – Single & Multi-level Sort
- Filtering Data – AutoFilter & Custom Filters

POST-CLASS (Assignment)

- Formula Exercise: Build a grade calculator using nested IF – assign A/B/C/D/F based on marks
- Text Exercise: Clean a messy dataset – fix casing, remove extra spaces, split names
- Date Exercise: Calculate each employee's age and years of service from DOB and join date
- VLOOKUP Exercise: Create a product lookup – enter a product ID and retrieve its name and price
- XLOOKUP Exercise: Replace VLOOKUP with XLOOKUP and compare results
- COUNTIF/SUMIF Exercise: From a sales dataset, count sales by region and sum revenue by product
- Table Exercise: Convert a dataset to an Excel Table and use structured references in formulas
- Mini Project: Employee Salary Calculator – use IF, VLOOKUP, and date functions to compute tax, bonus, and net pay

DAY 3 — Data Visualization – Charts, Conditional Formatting & PivotTables

Objective: *Visualize data meaningfully with charts, highlight key insights using Conditional Formatting, and summarize large datasets using PivotTables and PivotCharts.*

PRE-CLASS (Self Study)	IN-CLASS (Topics We Will Cover)	POST-CLASS (Assignment)
• Read: Types of Excel Charts and when to use them • Download a sample sales dataset (e.g., Superstore dataset) for practice • Read: What is Conditional Formatting in Excel? • Get an overview on Sparklines in Excel	• Introduction to Data Visualization in Excel • Chart Types – Column, Bar, Line, Pie, Doughnut, Area • Advanced Charts – Scatter Plot, Bubble Chart, Combo Chart • Statistical Charts – Histogram, Box & Whisker, Waterfall, Funnel • Creating and Formatting Charts – Titles, Axes, Legends, Data Labels • Chart Styles and Color Themes • Sparklines – Line, Column, Win/Loss in cells • Conditional Formatting – Highlight Cell Rules • Conditional Formatting – Top/Bottom Rules, Data Bars, Color Scales, Icon Sets • Custom Conditional Formatting with Formulas • Introduction to PivotTables – What & Why • Creating a PivotTable from scratch • PivotTable Fields – Rows, Columns, Values, Filters • PivotTable Calculations – Sum, Count, Average, % of Total • Grouping in PivotTables – Dates by Month/Quarter/Year • Slicers and Timelines for PivotTable Filtering • PivotCharts – Creating charts directly from PivotTables • Refreshing PivotTable Data	• Chart Exercise: From a sales dataset, create a Column chart for monthly revenue, a Pie chart for product share, and a Line chart for trend • Combo Chart: Plot revenue (bars) and profit margin % (line) on the same chart • Conditional Formatting: Highlight top 10 sales in green, bottom 10 in red, apply data bars to quantities • Sparklines: Add sparklines in a summary table to show sales trends per product per month • PivotTable Exercise: From a 500-row sales dataset, build a PivotTable showing revenue by region and category • PivotTable Exercise: Add a calculated field to compute profit margin % in the PivotTable • Slicer Exercise: Add slicers for Region and Year, connect them to both a PivotTable and PivotChart • Mini Project: Executive Sales Dashboard – PivotTable + PivotChart + Slicers + Conditional Formatting on one sheet

DAY 4 — Advanced Excel – Data Validation, Power Query & Dynamic Arrays

Objective: *Control data entry with validation, automate data import and transformation with Power Query, and use modern dynamic array functions for powerful analysis.*

PRE-CLASS (Self Study)	IN-CLASS (Topics We Will Cover)	POST-CLASS (Assignment)
• Read: What is Data Validation in Excel? • Read: Power Query Introduction • Read: What are Dynamic Array Functions in Excel (Microsoft Docs)? • Get an overview on Excel's LAMBDA function • Read: What is Power Pivot and the Data Model?	• Data Validation – Whole Number, Decimal, List (Dropdown), Date Range • Data Validation – Custom Formula-based validation • Input Messages and Error Alerts • Dependent Dropdown Lists using INDIRECT() • Named Ranges – Creating, Managing, Using in formulas • Dynamic Array Functions – FILTER(), SORT(), SORTBY(), UNIQUE() • Dynamic Array Functions – SEQUENCE(), RANDARRAY() • Lookup Arrays – CHOOSECOLS(), CHOOSEROWS(), TAKE(), DROP() • Spill Range References with # operator • Text Functions (Modern) – TEXTSPLIT(), TEXTBEFORE(), TEXTAFTER() • Introduction to Power Query (Get & Transform Data) • Power Query: Importing data from CSV, Excel, Web • Power Query: Transformations – Filter, Sort, Remove Duplicates, Split Column • Power Query: Add Custom Columns, Replace Values, Change Data Types • Power Query: Merge Queries (like VLOOKUP/JOIN) • Power Query: Append Queries (stacking tables) • Refreshing Power Query connections • Introduction to LAMBDA() – Custom reusable functions • LET() Function – Variable declarations in formulas • Introduction to Power Pivot & Data Model (overview)	• Data Validation: Create an Employee Entry Form with dropdowns for Department, Role, and date validation for joining date • Dependent Dropdown: Build a Country → State → City dependent dropdown system • FILTER Exercise: From a product list, dynamically filter all items in a chosen category priced above a threshold • SORT/UNIQUE Exercise: Extract a unique sorted list of customers from a sales log • SEQUENCE Exercise: Build a dynamic calendar for any month using SEQUENCE • Power Query Exercise: Import a messy CSV, clean it (remove blanks, fix types, split columns), and load to a sheet • Power Query Merge: Join two tables (Orders + Products) using Merge Queries to enrich order data • LAMBDA Exercise: Create a reusable LAMBDA to calculate BMI from height and weight • LET Exercise: Rewrite a complex nested formula using LET for readability • Mini Project: Automated Data Pipeline – Power Query imports raw CSV, cleans it, and feeds a summary PivotTable that refreshes with one click

DAY 5 — Automation, Macros, VBA & Capstone Dashboard

Objective: *Automate repetitive tasks with Macros and VBA, protect workbooks, build a professional dashboard, and apply best practices for real-world Excel projects.*

PRE-CLASS (Self Study)	IN-CLASS (Topics We Will Cover)	POST-CLASS (Assignment)
• Read: What is a Macro in Excel? • Read: Record a Macro in Excel – Beginner (YouTube – 10 min) • Read: Introduction to VBA (Visual Basic for Applications) – overview article • Read: Excel Workbook Protection basics • Get an overview on Excel Add-ins	• Introduction to Macros – What & Why • Recording a Macro – Absolute vs Relative References • Running and Editing Macros • Assigning Macros to Buttons • Introduction to VBA Editor (Alt+F11) • VBA Basics – Variables, Data Types (Dim, Dim As String/Integer/Double) • VBA Control Flow – If/ElseIf/Else, Select Case • VBA Loops – For Next, For Each, Do While, Do Until • VBA Sub Procedures and Functions • Working with Ranges in VBA – Range(), Cells(), ActiveCell • Working with Worksheets and Workbooks in VBA • VBA MsgBox and InputBox • VBA Error Handling – On Error GoTo, Resume Next • Creating User-Defined Functions (UDFs) with VBA • Workbook & Sheet Protection – Passwords, Allow Edit Ranges • Hiding Sheets, Locking Cells, Protecting Formulas • Form Controls – Button, ComboBox, CheckBox • Excel Dashboard Design Principles • Building a Dynamic Dashboard – Layout, Navigation Buttons, KPI Cards • Excel Best Practices – Naming conventions, documentation, error-proofing • Sharing & Collaboration – Share Workbooks, Track Changes, Comments	• Macro Exercise: Record a macro to format a report (apply header styling, borders, and freeze panes) in one click • VBA Exercise: Write a Sub to loop through a list of names and insert a greeting in the adjacent column • VBA Exercise: Build a form using InputBox to collect student data and append it to a table • UDF Exercise: Create a custom VBA function =TaxAmount(salary, rate) that calculates tax • VBA Error Handling: Add error handling to your file-reading macro so it alerts the user if the file is missing • Protection Exercise: Lock all formula cells in a workbook, protect the sheet with a password, and allow only data-entry cells to be editable • Form Controls: Add a dropdown ComboBox and a Button to a summary sheet to filter and refresh a table • CAPSTONE PROJECT: Build a Complete Business Intelligence Dashboard • → Import raw sales CSV using Power Query • → Clean and transform data (remove nulls, fix types, add calculated columns) • → Build PivotTables for revenue by region, product, and time • → Create 5 charts (trend line, category bar, regional map, top-10, KPI gauge) • → Add Slicers and Timelines for interactivity • → Add a Macro button to refresh all data in one click • → Protect formulas and structure, add navigation buttons between sheets • → Present the dashboard with a 5-slide summary of key insights

Quick Reference —Overview

Day	Topic	Focus	Key Skills Gained
Day 1	Setup & Fundamentals	Interface, Data Entry, Formatting, Basic Formulas	Navigate Excel, format data, use SUM/AVERAGE
Day 2	Formulas, Functions & Data Management	IF/IFS, Lookup, Text, Date, Tables, Sort & Filter	Build dynamic calculations, look up and clean data
Day 3	Charts, Conditional Formatting & PivotTables	Charts, Sparklines, CF, PivotTables, Slicers	Visualize data, create dashboards, summarize with PivotTables
Day 4	Advanced Excel & Power Query	Dynamic Arrays, Data Validation, Power Query, LAMBDA	Automate data import, use modern array functions
Day 5	Macros, VBA & Capstone Dashboard	Macros, VBA, Protection, Dashboard, Best Practices	Automate tasks, build a full BI dashboard